



SAYAN DAS



MT25041, Email: sayan25041@iiitd.ac.in

DOB: January 20, 2002

Address: Habibpur Purba, Midnapore, West Bengal, 721101

Education

Indraprastha Institute of Information Technology, Delhi M.Tech(CSE) 2025 – Present	CGPA: 7.76 (till 2 nd Semester)
Ramkrishna Mahato Government Engineering College, West Bengal B.Tech(CSE) 2021 – 2025	CGPA: 8.83
Midnapore Collegiate School (Boys), Midnapore (West Bengal) WBCHSE 2018 – 2020	Percentage: 93.4
Sarada Vidyamandir, Midnapore (West Bengal) WBBSE 2018	Percentage: 93.14

Skills

Expertise Area	ML (NLP, Gen AI, Computer Vision), Container Infrastructure, Computer Networks
Programming Languages	C, C++, Python
Tools and Technology	Git, Linux, Docker, Kubernetes, MySQL, HTSIM, ns-3
Frameworks	PyTorch, TensorFlow, Scikit-Learn, Pandas
Technical Electives	Natural Language Processing, Machine Learning, Usable Security and Privacy, Information Integration and Applications

Work Experience

Software Engineering Intern, Foodoscope	(May,26 – Present)
Chatbot Development: Led the team in designing containerized RAG and semantic search pipelines; integrated local SQLite databases to reduce response latency, and enforced read-only SQL guardrails to reduce hallucinations and prevent unauthorized database modifications/data leakage. Backend Development: Developed RESTful APIs for FlavourDB using MongoDB aggregation, enabling mapping and retrieval of molecules associated with foods.	

Projects

Enhancing Code-Mixed Speech Recognition using Whisper Fine-Tuning and LLM-Based Contextual Rescoring

(Jan,26 – May,26)

Team Size-2

Guide: Dr. Md Shad Akhtar

- Built a **Bengali-English code-switched ASR** pipeline by **fine-tuning the Whisper** decoder with **prompt conditioning** and **LLM-based n-best rescoring** (GPT-2, BLOOM, Qwen2.5), **cutting WER from 195% (zero-shot) to 39.6%**.
- Extended the base method (**CLEAR, ICNLSP 2025**) with **ASR-LLM score interpolation, MBR decoding, and a LoRA domain-adapted rescorer (4% relative WER gain)**; built with **PyTorch, Hugging Face Transformers**

Mini-GPT: From-Scratch Transformer with Edge-Efficiency Benchmarking

(Feb,26 – Apr,26)

Team Size-2

Guide: Dr. Md Shad Akhtar

- Built a GPT-style Transformer from scratch, including attention, RoPE, and **RMSNorm**, and verified it against PyTorch's own implementation. The modern version beat the vanilla one by 4.3% on validation perplexity at the same size. Benchmarked **FP32, FP16, and INT8** versions plus a distilled model to measure real speed-vs-quality trade-offs.
- **PyTorch, SentencePiece, scikit-learn, LoRA, mixed-precision training.**

Ultra Ethernet Transport (UET) Simulator for High-Performance Computing and AI Workloads

(Jan,26 – Jun,26)

Guide: Prof. Rinku Shah

- Extended the **ns-3 simulator in C++17** to implement **Ultra Ethernet Transport (UET)**, a next-gen protocol for **AI/HPC data centers** — including **packet delivery, reliability logic, and full protocol state-machine implementation** per the official spec, using **RDMA/RoCEv2**.
- Used **C++17, ns-3** to showcase networking, communication protocols, and system-level programming.

Performance Benchmarking of Kubernetes CNI Plugins

(Jan,26-Jun,26)

Guide: Prof. Rinku Shah

Team Size-4

- Benchmarked Kubernetes **CNI plugins (Flannel-VXLAN, Calico-BGP/VXLAN, Cilium-eBPF)** across a **multi-node cluster**, measuring throughput, latency, CPU, and memory to identify the most efficient networking backend.
- Used **Kubernetes, Linux, Bash, Python, iPerf3**, and **container**, networking tools

Positions of Responsibility

- Teaching Assistant - Introduction to Programming (Aug,25 – Dec,25)
- Teaching Assistant - Network Security (Jan,26 – May,26)

Awards and Achievements

- GATE CSE & GATE DA Qualified in 2024 and 2025

Interests and Hobbies

- Drawing

Declaration: The above information is correct to the best of my knowledge.

Sayan Das

Date: Sep 8, 2026